主修---Programming Part 02 (19-23)

Name: _____ ( ) Class: _____

2019

Peter designs a program using an algorithm with an array A, as shown below:

```
N ← 6
For I from 1 to N do
    A[I] ← 1 - A[I]
```

(a) (i) Suppose that the initial content of A is:

A[6]	A[5]	A[4]	A[3]	A[2]	A[1]
0	0	1	1	0	0

What is the content of A after executing the algorithm?

A[6]	A[5]	A[4]	A[3]	A[2]	A[1]

(2 marks)

(ii) Suppose that the content of A after executing the algorithm is:

A[6]	A[5]	A[4]	A[3]	A[2]	A[1]
1	0	1	0	1	0

What is the initial content of A?

A[6]	A[5]	A[4]	A[3]	A[2]	A[1]

(1 mark)

Peter modifies the algorithm below:

```
N ← 6
K ← 2
For I from 1 to N do
   If K = 1 then
         A[I] ← 1 - A[I]
   If A[I] = 1 then
         K ← 1
```

(b) (i) Suppose that the initial content of `A` is:

A[6]	A[5]	A[4]	A[3]	A[2]	A[1]
0	0	1	1	0	0

(1) What is the content of `A` after executing the algorithm once?

A[6]	A[5]	A[4]	A[3]	A[2]	A[1]

(2) What is the content of `A` after executing the algorithm once again? (2 marks)

A[6]	A[5]	A[4]	A[3]	A[2]	A[1]

(ii) Suppose that the content of `A` after executing the algorithm is: (1 mark)

A[6]	A[5]	A[4]	A[3]	A[2]	A[1]
1	0	1	0	1	0

What is the initial content of `A`?

A[6]	A[5]	A[4]	A[3]	A[2]	A[1]
0	1	0			

(2 marks)

(iii) Write down the initial content of `A` such that the content remains unchanged after executing the algorithm.

A[6]	A[5]	A[4]	A[3]	A[2]	A[1]
0	0	0			

(1 mark)

(c) Peter will execute the program millions of times with `N >= 64` on his desktop computer and wants to improve the performance of the program execution significantly.

(i) Increasing the size of RAM does not help. Why not?

_____

_____

(1 mark)

(ii) Suggest a hardware component that should be upgraded.

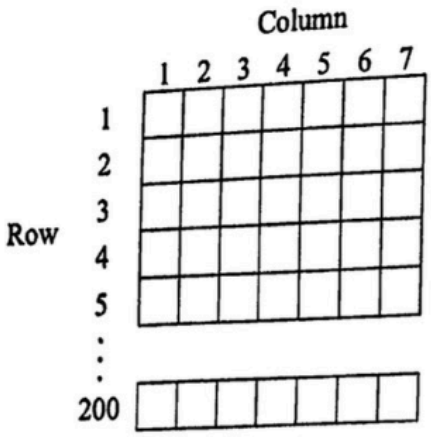
_____

(1 mark)

		Marks
5. (a) (i)	110011	2
(ii)	010101	1
(b) (i)	(1) 110100	2
	(2) 001100	1
(ii)	110	2
(iii)	000	1
(c) (i)	The algorithm does not need much memory to be executed.	1
(ii)	CPU	1

2020

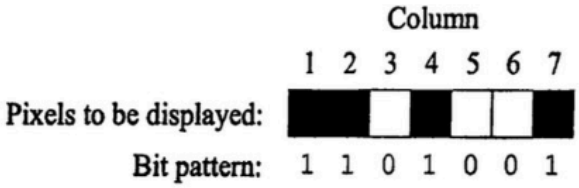
5. A computer controls a display consisting of 200 × 7 pixels, as shown below:



A white pixel and a black pixel are represented by '0' and '1' respectively. Method 1 and Method 2 are methods that use bit patterns to represent pixels in the display.

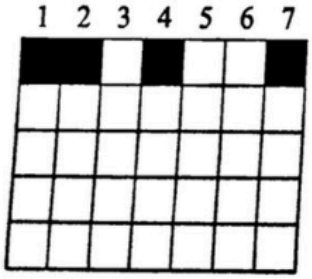
Method 1

A bit pattern consists of 8 bits. The first 7 bits represent a row of pixels and the 8th bit is always 1. The following example shows how `11010011` represents a row of pixels:



(a) Complete the display below by drawing pixels to represent the following five bit patterns, in order.

`11010011` `10000011` `10000011` `10000011` `01111101`

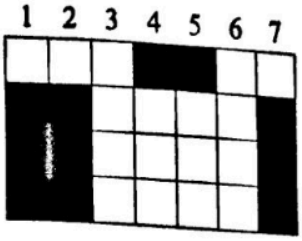


(2 marks)

Method 2

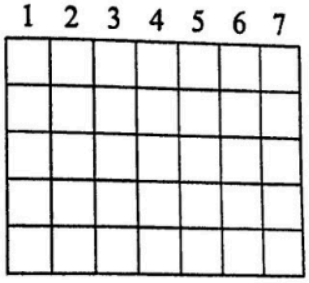
Similar to Method 1, if the 8th bit of a bit pattern is 1, the first 7 bits represent a row of pixels. If the 8th bit of a bit pattern is 0, the first 7 bits represent the number of occurrences (in binary notation) of the row of pixels represented by the next bit pattern.

For example, 00011001 00000110 11000011 represent



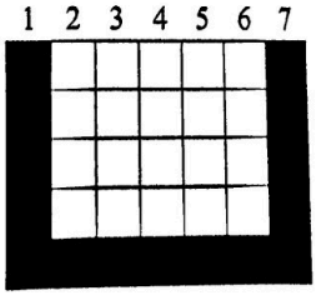
(b) (i) Complete the display below by drawing the pixels represented by the following four bit patterns in order.

11111111 00000110 11010011 11111111



(2 marks)

(ii) Write down the three bit patterns for representing the following display:



Bit patterns: _____ _____ _____

(2 marks)

(iii) What is the maximum number of rows of pixels represented by two bit patterns?

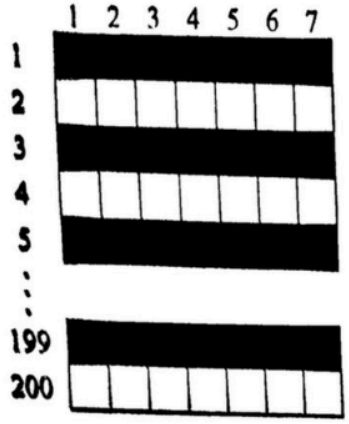
_____

(2 mark

(c) (i) In which type of memory, RAM or ROM, should the information about the pixels in the display be stored? Give a reason to support your choice.

(1 mark)

(ii) The following display consists of alternate rows of black pixels and white pixels. Compare the memory sizes needed for the following display represented by Method 1 and Method 2 respectively. Explain briefly.

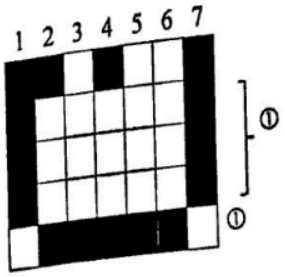


(2 marks)

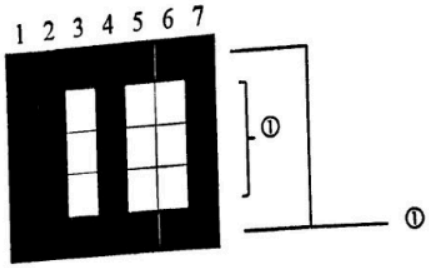
(iii) In general, which method, Method 1 or Method 2, needs fewer computer resources? Explain briefly.

(2 marks)

5. (a)



(b) (i)



(ii) 00001000 10000011 11111111

① ①

(iii) 127

(c) (i) RAM should be used as it is volatile and its content can be over-written.

(ii) Memory size needed by Method 1 <= Memory size needed by Method 2
Method 1 = 200 bytes, Method 2 = 200 bytes or 400 bytes
Reasonable explanation

(iii) Method 2 needs fewer computer resources.
Whenever there are any three or more repeated rows to be displayed, only two-bit patterns are needed. It saves storage space to represent the content.

Or

Method 1 needs fewer computer resources.
Method 1 is simple in structure and saves the computational power/time.

2021

Peter designs an algorithm to encrypt an array of binary digits, A, as shown below:

```
I ← 5
repeat
       I ← I - 1
       A[I] ← 1 - A[I]
until A[I] = 1
```

(a) (i) Suppose that the initial content of A is:

A[1]	A[2]	A[3]	A[4]
0	0	1	1

What is the content of A after executing the algorithm?

A[1]	A[2]	A[3]	A[4]

(2 marks)

(ii) Suppose that the content of A after executing the algorithm is:

A[1]	A[2]	A[3]	A[4]
1	0	1	0

What is the initial content of A?

A[1]	A[2]	A[3]	A[4]

(2 marks)

(iii) The algorithm cannot terminate properly if all entries in the initial contents of A are 1. Revise the algorithm such that it can terminate properly.

```
I ← 5
repeat
       I ← I - 1
       A[I] ← 1 - A[I]
until (A[I] = 1)  [          ]  [                    ]
```

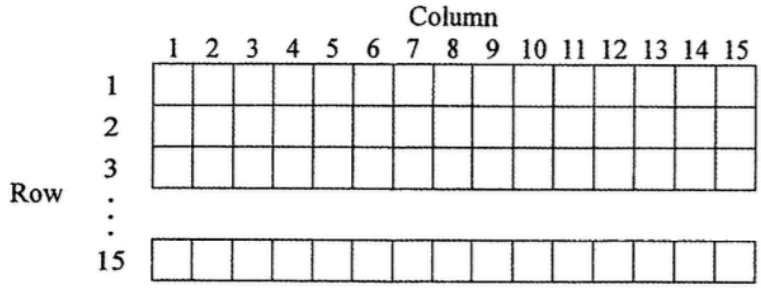
(2 marks)

(b) Peter rewrites the algorithm in (a)(iii) as an alternative algorithm. Complete the algorithm below.

```
Flag ← 1
for I from [      ] down to 1 do
    if Flag = [      ] then
          A[I] ← 1 - A[I]
    if A[I] = [      ] then
          Flag ← 0
```

(3 marks)

A display consists of 15 × 15 pixels, as shown below:



Each pixel displays a colour represented by a 2-bit code, as shown below:

Colour	Short form	2-bit code
Red	R	00
Green	G	01
Blue	B	10
Yellow	Y	11

(a) How many bits are required for representing the pixels on the display? Show your calculation.

(2 marks)

A computer applies an encoding scheme that each row on the display is represented by one or more bit patterns. Each bit pattern consists of 6 bits. The first 2 bits represent a colour, and the remaining 4 bits represent the number of consecutive pixels (in binary notation) of the colour.

For example, the bit pattern `110100` represents 4 consecutive yellow pixels:

`11` (Y) `0100` (4)

(b) Write down the pixels that the bit pattern `100001` represents.

Colour:_____ Number of pixels:_____

(2 marks)

For example, the two bit patterns `110100` and `001011` represent a row of 15 pixels:

Y	Y	Y	Y	R	R	R	R	R	R	R	R	R	R	R

(c) (i) Write down the pixels represented by the following three bit patterns.

`010100` `100001` `111010`

(2 marks)

(ii) Write down the bit patterns for representing the following pixels in a row.

B	B	Y	B	B	B	B	B	B	B	B	B	B	B	B

(2 marks)

(iii) With the encoding scheme, how many bits are required for representing a row of 15 pixels in each of the following cases?

(1) All pixels are red.

(1 mark)

(2) No two adjacent pixels are of the same colour.

(2 marks)

(iv) What computer resource should be considered for implementing the encoding scheme, other than the storage size? Explain briefly.

(2 marks)

(a) (i) 0100 1, 1
① ①

(ii) 1001 1, 1
① ①

(iii) OR (I = 1) 1, 1

(b) 4 1×3
1
1

(a) $15 \times 15 \times 2 = 450$ bits 2
- only expression ①
- only answer ②

(b) Blue/B 1 1, 1

(c) (i)

G	G	G	G	B	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

2
Any one code ①

(ii) 100010 110001 101100 Any one ① 2

(iii) (1) 6 1

(2) $15 \times 6 = 90$ 2
- only expression ①
- only answer ②

(iv) The computation power (CPU) ① should be considered because the encoding scheme involves extra computation for representation. ① 1+1

2022

(d) John considers encoding symbols into bit patterns in the application. The following table shows the bit patterns of symbols using two different methods.

Symbol	Bit pattern	
	Method 1	Method 2
☺	10	1
☹	01	01
😐	00	000
☹	11	001

(i) Encode ☺ 😐 ☹ ☹ using Method 1: _____

(1 mark)

(ii) Decode '001011' using Method 2: _____

(1 mark)

(iii) Find the number of bits required for encoding the symbols in each row below.

	Method 1	Method 2
☺ ☺ ☺ ☹ ☹ ☹		
☹ ☹ ☹ 😐 😐 😐		

(2 marks)

(iv) Under what scenario will Method 2 be better than Method 1?

_____

_____

(1 mark)

Mary designs a subprogram with an array A. She considers algorithms ALG1, ALG2 and ALG3.

ALG1

Line number	Content
1	B ← 0
2	for i from 1 to N do
3	if A[i] = 1 then
4	B ← B + 1
5	if B >= N/2 then
6	ch ← 'X'
7	else
8	ch ← 'Y'

(a) Suppose that N = 6. Execute ALG1 with the following initial content of A:

A[1]	A[2]	A[3]	A[4]	A[5]	A[6]
1	0	1	0	0	0

Write down the final values of B and ch.

B:_____ ch:_____

(2 marks)

(b) Suppose that N = 6. Execute ALG1 with the following initial content of A:

A[1]	A[2]	A[3]	A[4]	A[5]	A[6]
1	1	1	1	0	0

How many times will Line 3 be executed? _____ (1 mark)

ALG2

Line number	Content
1	i ← 1
2	B ← 0
3	while (i <= N) AND (B < N/2) do
4	if A[i] = 1 then
5	B ← B + 1
6	i ← i + 1
7	if B >= N/2 then
8	ch ← 'X'
9	else
10	ch ← 'Y'

(c) Suppose that N = 6. Execute ALG2 with the initial content of A:

A[1]	A[2]	A[3]	A[4]	A[5]	A[6]
1	1	1	1	0	0

(i) How many times will Line 4 be executed? _____ (1 mark)

(ii) ALG2 can be more efficient than ALG1. Why?

(1 mark)

(d) Complete the following initial content of `A` such that the number of executions of the statement '`if A[i] = 1 then`' in both ALG1 and ALG2 would be the same.

A[1]	A[2]	A[3]	A[4]	A[5]	A[6]
1	1				

(2 marks)

(e) ALG1 and ALG3 below should always produce the same result. Complete Line 3 below.

ALG3

Line number	Content
1	`B ← 0`
2	`for i from 1 to N do`
3	`    B ← B + A [ _____ ]`
4	`if B >= N/2 then`
5	`    ch ← 'X'`
6	`else`
7	`    ch ← 'Y'`

(1 mark)

(f) The value of `N` is very large. Which algorithm, ALG1 or ALG3, is more efficient? Explain briefly.

(2 marks)

(g) A computer executes Mary's subprogram. State **two** components in the CPU that are involved in the fetch-decode-execute cycle, other than IR (instruction register).

(2 marks)

(d) (i) `10 00 11 11` 1

(ii) ☹ 😐 ☺ 1

(iii)

	Method 1	Method 2
☺☺☺ ☹☹☹	12	12
☹☹☹ 😐😐😐	12	18

2

(iv) The message contains more ☺. 1

(a) `B = 2` `ch = Y` 1,1

(b) 6 1

(c) (i) 3 1

(ii) ALG2 is more efficient as the number of comparisons / executions with the conditional statement on Line 3 is fewer. 1

(d) `110000` or `110001` 2

(e) `i` 1

(f) ALG3. Fewer instructions have to be executed as the conditional statement "`if A[i] = 1 then`" is removed. 2*

(g) MDR (memory data register) / MAR (memory address register) / PC (program counter) / ALU (arithmetic logic unit) / CU (control unit) / SR (status register) 1×2

2023

Eva writes a subprogram with the following pseudocode to sort an array `A`.

Line number	Content
1	`i ← 1`
2	`j ← 2`
3	`while i < 5 do`
4	`    if A[i] >= A[i-1] then`
5	`        i ← j`
6	`        j ← j + 1`
7	`    else`
8	`        swap the contents of A[i] and A[i-1]`
9	`        i ← i - 1`
10	`        if i = 0 then`
11	`            i ← 1`

(a) Suppose that the initial content of `A` is:

A[0]	A[1]	A[2]	A[3]	A[4]
4	3	1	8	6

(i) `TEMP` is a variable. Complete the following pseudocode to represent Line 8.

```
TEMP ← A[i]
```

_____

_____

(1 mark)

(ii) What are the contents of `A`, `i` and `j` after completing the first, second and third iterations of the loop (Lines 3 to 11)?

First iteration:

A[0]	A[1]	A[2]	A[3]	A[4]

i: _____ j: _____

Second iteration:

A[0]	A[1]	A[2]	A[3]	A[4]

i: _____ j: _____

Third iteration:

A[0]	A[1]	A[2]	A[3]	A[4]

i: _____ j: _____

(5 marks)

(b) Set 1 and Set 2 below are two sets of values as the initial content of `A`. How many times will Line 8 be executed for each set?

Set 1

A[0]	A[1]	A[2]	A[3]	A[4]
2	3	5	7	9

Set 2

A[0]	A[1]	A[2]	A[3]	A[4]
9	7	5	3	2

Set 1: _____ Set 2: _____

(3 marks)

(c) On Line 10, there will be no changes in the result of the subprogram if '`i = 0`' is changed to '`i <= 0`'. Why?

_____

_____

(1 mark)

(d) Below are three sets of test data as the initial content of `A`. Choose **two** sets for testing the subprogram and explain briefly.

Test 1: `-3, -1, 4, 0, 7`
Test 2: `1, 2, 3, 4, 5`
Test 3: `1, 1, 3, 3, 5`

_____

_____

_____

_____

(2 marks)

(a) (i) `A[i]←A[i-1]`
`A[i-1]← TEMP` 1

(ii) First iteration:

A[0]	A[1]	A[2]	A[3]	A[4]	
3	4	1	8	6	1

`i:1` `j:2` 1

Second iteration:

A[0]	A[1]	A[2]	A[3]	A[4]	
3	4	1	8	6	1

`i:2` `j:3` 1

Third iteration:

A[0]	A[1]	A[2]	A[3]	A[4]	
3	1	4	8	6	1

`i:1` `j:3`

(b) Set 1: 0 Set 2: 10 1, 2

(c) `i` will never be smaller than 0. 1

(d) Test 1: It contains negative values / zero.
Test 2: It is sorted / unsorted / normal / standard data.
Test 3: It contains duplicate values. 1×2